Arul Rhik Mazumder

U.S. Citizen

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Research interests. What useful quantum computation really costs: proving how many expensive operations (T gates) and extra qubits a circuit needs, designing cheaper algorithms for error-corrected quantum computers, and finding where quantum algorithms genuinely beat classical ones.

Education

Imperial College London **Sep. 2026 – Sep. 2027 (expected)**

MSc, Physics with Quantum Dynamics *London, UK*

- Single-author preprint (submitted): randomized Toffoli gates that need no extra qubits once the gate is large enough, with a T-count within a factor of eight of the fewest possible among measurement-free randomized circuits.

Carnegie Mellon University **Aug. 2023 – May 2026**

B.S. in Computer Science; ML and Algorithms & Complexity concentrations; University Honors *Pittsburgh, PA*

- Coursework: Quantum Computing, Complexity Theory, Learning Theory, Randomized Algorithms, Convex Optimization, Parallel Computer Architecture.

Research Experience

Institute for Quantum Computing (IQC), University of Waterloo **May 2026 – Aug. 2026**

Undergraduate Researcher (Mitacs Globalink Scholar), hosted by Michele Mosca and Vlad Gheorghiu *Waterloo, ON, Canada*

- **Single-author result** (arXiv:2610.01024): proved that no circuit without internal Hadamard gates can implement a layer of m non-overlapping Toffoli-type (CCZ) gates with fewer T gates than Campbell and Howard's $6m + 1$ construction, using a lower bound, computable in polynomial time, for a broad class of gates (diagonal Clifford+T).
- On the output of two public circuit optimizers (PyZX, TODD), the bound proves 193 of 311 circuit pieces already optimal, against 113 for stabilizer nullity, the standard prior bound.

Magic State Labs **Aug. 2026 – Oct. 2026**

Research Consultant, with Pradeep Niroula *Remote*

- Wrote criteria for checking whether a problem is a sound target for decoded quantum interferometry (DQI), a recent quantum optimization algorithm, and used them to steer an AI-agent search that found two new candidate applications (arXiv:2610.06753).

Caltech Institute for Quantum Information and Matter (IQIM) **May 2025 – Aug. 2025**

SURF Undergraduate Researcher, Preskill group, with Samson Wang and James Watson (Google Quantum AI) *Pasadena, CA*

- First-author paper (arXiv:2608.13862): quantum algorithms for computing functions of matrices on early error-corrected hardware, with costs that shrink when the terms of the simulated system nearly commute.

BlueQubit **Oct. 2025 – Present**

Part-time Researcher, AI and Core teams *Los Angeles, CA (remote)*

- First-author benchmark of GPU-accelerated quantum simulators against commercial and open-source platforms (arXiv:2607.09882); the results informed product decisions and cost estimators.

Quantum Technologies Group, Carnegie Mellon University **May 2023 – Jun. 2025**

Researcher with Sridhar Tayur *Pittsburgh, PA*

- Wrote a tutorial on solving QUBO problems with QAOA and quantum annealing, with five open notebooks (*Tutorials in Operations Research*, INFORMS 2025).
- Implemented semidefinite-programming formulations for entanglement detection (CVXPY, PICOS, Gurobi).

Algorithms Group, George Mason University **May 2022 – Jun. 2025**

Researcher with Fei Li *Remote*

- Three papers on quantum and hybrid algorithms: string matching in Qiskit (AISS 2022); a hybrid D-Wave algorithm for the traveling salesman problem that beat pure classical and pure quantum baselines (IPDPS 2023); and shipment rerouting with up to 1,000 shipments on real transport networks (JSSR 2024).

Publications

Preprints

1. **A. R. Mazumder**. "Exact T-counts of CCZ Layers from an Isotropy Bound." arXiv:2610.01024, 2026.
2. **A. R. Mazumder**. "Ancilla-free Randomized Toffoli Gates and the T-count of Boolean Oracles." arXiv, submitted 2026.
3. **A. R. Mazumder**, P. Niroula. "Discovering New Problems for Decoded Quantum Interferometry." arXiv:2610.06753, 2026.
4. **A. R. Mazumder**, J. Watson, S. Wang. "Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling." arXiv:2608.13862, 2026.

5. **A. R. Mazumder**, H. Karapeytan, M. Z. Mullath, R. Tanin, H. Gharibyan, H. Tepanyan. "Benchmarking Zero-Setup Quantum Circuit Simulators." arXiv:2607.09882, 2026.
6. F. Li, **A. R. Mazumder**, M. Zhao. "Quantum Annealing Approaches to Shipment Rerouting Problems." arXiv:2501.05624, 2025.

Conference and journal papers

1. **A. R. Mazumder**, E. Towe. "An End-to-End Hybrid Quantum–Classical Sampling Workflow for Discrete Markov Random Fields." *IEEE QCE*, 2026.
2. **A. R. Mazumder**, S. R. Mazumder. "Depth-Efficient Quantum Topological Data Analysis for Regime-Specific Detection of Financial Stress." *IEEE QCE*, 2026.
3. **A. R. Mazumder**, S. Tayur. "Five Starter Problems: Solving Quadratic Unconstrained Binary Optimization Models on Quantum Computers." *Tutorials in Operations Research*, INFORMS, 2025, pp. 145–183.
4. M. Zhao, **A. R. Mazumder**, F. Li. "Improved Classical and Quantum Algorithms for the Shipment Rerouting Problems." *J. Student-Scientists' Research*, 2024.
5. **A. R. Mazumder**, A. Sen, U. Sen. "Benchmarking Metaheuristic-Integrated QAOA Against Quantum Annealing." *LNNS*, 2024.
6. F. Li, **A. R. Mazumder**. "An Adaptive Hybrid Quantum Algorithm for the Metric Traveling Salesman Problem." *IPDPS*, 2023.
7. M. Gao, R. Huang, **A. R. Mazumder**, F. Li. "Comparisons of Classic and Quantum String Matching Algorithms." *AISS*, 2022.

Awards and Honors

- **Mitacs Globalink Research Award** and **IQC Undergraduate Research Award** (2026); **Caltech SURF Fellow** (2025); **IEEE QCE Student Travel Grant** (2026).
- **Duke STAQ NSF-funded Invited Scholar** (2025): selected for a PhD-level week on quantum simulation, error correction and hardware architecture.
- **YQuantum** (2025, Yale): Yale Quantum Institute Grand Prize; **MIT iQuHACK** (2026): 2nd of 90+ teams, NVIDIA track.
- **Citadel/Correlation One Datathon** (2024): 2nd of 150 invited from 3,000+ applicants; **HackCMU** (2023): prize for an AI waste sorter piloted with CMU Dining.
- **Regeneron Science Talent Search Scholar** (2023), top 300 of 2,000+ nationally. **Olympiads:** USA(J)MO qualifier, USAPhO Bronze, USACO Gold, USNCO National Finalist.

Leadership and Outreach

CMU Quantum **Sep. 2024 – May 2026**
Founder and President *Pittsburgh, PA*

- Founded CMU's first student quantum organization and grew it to 200+ members and 10+ student leaders; secured sponsorship from General Motors and Boeing.
- Wrote teaching materials; ran workshops with IBM, IonQ and Quantinuum; led hackathon tracks and organized sponsored research trips.

BlueQubit **Oct. 2025 – Present**
Hackathon Lead *Remote*

- Mentor students and judge the BlueQubit track at student quantum hackathons; write technical blog posts and tutorials.

Talks and Posters

- **QRE 2026** and **UnitaryCon 2026**, Toronto — posters, "Certified T-count floors for Toffoli networks and reversible circuits."
- **TQC 2026** — poster, "Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling" (PDF).
- **Caltech IQIM SURF Seminar**, Aug. 2025, and **Quantum Leap Career Nexus**, UMD, Nov. 2025 — talks, "Early Fault-Tolerant Quantum Algorithms for Matrix Functions via Trotter Extrapolation" (slides).
- **INFORMS Annual Meeting 2025** — talk on solving QUBO models on quantum computers.

Technical Skills

Languages and proof tools: Python, Julia, C++, Lean 4
Quantum and optimization: Qiskit, PennyLane, CUDA-Q, D-Wave Ocean; CVXPY, PICOS, Gurobi; CUDA, MPI